



BPMJ
9,5

652

A framework of dynamic CRM: linking marketing with information strategy

Chung-Hoon Park and Young-Gul Kim

*Graduate School of Management, KAIST, Dongdaemoon-Gu,
Seoul, South Korea*

Keywords Customer information, Customer relations, Customer loyalty

Abstract Committed customers are profitable to an organization for the long term. Customer commitment forms when a customer's expectation is satisfied and the customer realizes fair value from his/her relationship with the organization. From an organization's perspective, this value reflects customer equity, but from a customer's perspective, it represents the customer's perceived value of the relationship. In order to manage such a relationship successfully, it is necessary to support diverse customer information – such as of-the-customer, for-the-customer, and by-the-customer information. A customer information system (CIS) plays the role of boundary spanning that manages and distributes customer information. But the gap between marketing and IT strategy is a barrier in implementing a successful CIS. The CIS, which includes the database, communication channel, and decision model for relationship management, should be designed to facilitate the two-way customer relationship exchanges. This paper develops a framework of dynamic customer relationship management, suggests the information technology strategy to support the framework, and illustrates the applicability of such framework and strategy through a real business case.

Introduction

Effective customer relationship management (CRM) has become a strategic imperative for companies in virtually every business sector. Companies are moving closer to their customers, expending more effort in finding new ways to create value for their customers, and transforming the customer relationship into one of solution finding and partnering rather than one of selling and order taking (El Sawy and Bowles, 1997). Organizations will be more successful if they concentrate on obtaining and maintaining a share of each customer rather than a share of the entire market (Peppers and Rogers, 1995). It has been illustrated in practice that retaining an existing customer is more profitable than acquiring a new one (Reichheld and Sasser, 1990).

As for the key factors in a long-term relationship, marketing researchers have studied the effects of mutual information sharing exchange (Duncan and Moriarty, 1998), and communication effectiveness on building trust and commitment among relationship participants (Sharma and Patterson, 1999). Each relationship exchange incurs transactional and/or non-transactional data. Transactional data typically include sales amount, transaction time, place, and buyers while non-transactional data may include inquiries or feedback in the form of complaints or suggestions. Both transactional and non-transactional



data must be organized into an integrated customer data profile because such information is what makes customer interaction powerful (Wells *et al.*, 1999). Such integration is a foundation of simplifying customer support activities and reducing transaction costs so that the organization may not only differentiate its products but also offer lower prices (Goodhue *et al.*, 1992; Rangaswamy, 1993; El Sawy and Bowles, 1997).

However, CRM strategy means different things to different people (Winer, 2001). In the marketing area, much discussion has been devoted to relationship marketing strategy but a critical problem is that although a number of authors have contrasted relationship marketing with direct, database, loyalty, retention and transaction marketing, such discussions fail to clarify substantive issues (O'Malley and Tynan, 2000). For example, relationship marketing strategy has clarified customer lock-in (Barnes, 1994; Turnbull and Wilson, 1989; Palmer, 1995), customer retention (Berry, 1983), database marketing (Copulsky and Wolf, 1990; Treacy and Wiersema, 1993), and close relationships (Barnes, 1994, 1995). However, the first three approaches are criticized in that, all are lacking recognition of "mutuality and special status" (Rowe and Barnes, 1998, p. 284) and, therefore, are unlikely to result in close, personal, long-term relationships.

For the information technology domain, as key enabler of CRM, there has been active discussions on the implementation of relationship management package (Jutla *et al.*, 2001), customer information system (Mentzas, 1997; Wells *et al.*, 1999), customer database (El Sawy and Bowles, 1997), decision support application (Kohli *et al.*, 2001). Still, many companies have difficulty in implementing an effective CRM program because they allow software vendors to drive their approach to customer management, or retrofit a customer strategy to match the CRM technology they have purchased (Rigby *et al.*, 2002). Namely, most of them mistake CRM technology for a marketing strategy, forming a barrier in implementing a successful CRM project (Woodcock and Starkey, 2001).

In this paper, to help reduce such misconception, we develop a framework of dynamic CRM from a marketing perspective and suggest appropriate IT strategies to support the framework. We then illustrate the applicability of such a framework and strategy through a real business case.

Types of customers

Enabled by advanced information technologies, companies can now collect unprecedented amount and depth of data on their customers and turn them into information for their strategic business purposes. Here, the important issues are: to identify what kind of information they need; about whom they will collect this information; and how they will manage such information for future use. From an information management perspective, customer identification is a critical beginning point for CRM. When customers do not reveal their identity to the firm (like the cash-paying customers in retail stores or airline passengers

who are not enrolled in the mileage program), they are invisible to the firm, thus unmanageable. We define an identified customer as “a customer whose identification and contact information exists within the firm”. Once able to identify its customers, a firm can make its customer-base more loyal by collecting, processing, and applying customer profile and transaction data to create in-depth understanding of customer needs and provide fair value to all customers.

As the firms nurture their relationships with their identified customers, some of them will evolve to become, what we call, core customers. Core customers are typically a small group of customers who are truly loyal to the firm and create enormous value for the firm. Despite their significant contribution to the firm, identifying these core customers is not always as easy as it sounds. In fact, defining core customers is regarded as one of the most critical strategic processes (Blattberg and Deighton, 1996). We define the core customer of a firm as “an identified customer who accrues substantial value to the firm through high profit contribution, long term active relationship, or strong opinion leadership”.

It is not necessary to satisfy all three conditions to qualify as a core customer. Since the conditions are not mutually exclusive, some may satisfy two or all three conditions while others may satisfy only one of them. To identify core customers, firms need to collect not only transaction data (purchasing) but also various non-transaction data such as membership activity, new customer referral, or feedback idea for the firm.

Types of customer information

According to the content and interaction types, customer information can be classified into three types:

- (1) information of the customer;
- (2) information for the customer; and
- (3) information by the customer.

First, “of-the-customer” information includes personal and transaction data about a customer. It is the type of information most widely collected for CRM implementations. Firms obtain the personal data and are able to understand the customer’s sales volumes, profitability, purchasing patterns, frequency, preference, etc. For example, banks and credit card firms keep enormous amount of “of-the-customer” information in their database systems for opening, maintaining, and customer accounts billing and also to identify the most or least profitable customers. Database marketing, also called target marketing, is based on the strategic use of “of-the-customer” information.

Second, product, service and organizational information that are perceived useful by customers is referred to as “for-the-customer” information. This type of information is presented through diverse communication media so that

customers acquire and process it to make more informed decisions. Firms can provide such information by direct mail, automatic response system (ARS), or Internet home pages.

A framework of
dynamic CRM

The third type is “by-the-customer” information. This is the non-transactional customer feedback information that includes customer complaints, propositions, claims, A/S information, etc. Information of this type must be included in the expanded customer data profile because such information is what makes customer interactions powerful (Wells *et al.*, 1999). Since it contains customers’ direct complaints, needs and suggestions, this type of information can be applied to develop new products and services or improve critical business processes.

655

An integrated framework for dynamic CRM

Figure 1 shows the integrated customer relationship management framework based on customer information types along with relationship evolution stages. At the relationship initiation stage, firms identify customers by collecting and recording “of-the-customer” information. Registering customers into the firm’s membership or bonus point programs is a typical method for customer identification.

For the identified customers, firms can provide various “for-the-customer” type information – for example, newsletters, special promotion notice, bonus point status, order status, etc. – as well as customized service. Some of the identified customers, after a certain period of satisfactory relationship experience, eventually evolve into core customers who satisfy one or more of the criteria for core customer. As identified customers evolve into core customers, the firm enters the “expansion phase” in its CRM. In this phase, core customers actively participate in the two-way interactions with the firm and expand the firm’s customer base by word-of-mouth marketing. Feedback or suggestion from these core customers (“by-the-customer” information) may prove to be crucial for the firm to introduce new products, improve business processes, and satisfy customer needs. The boundary between the firm and its customers becomes transparent in this phase

Relationship value analysis

For effective customer relationship management, each customer relationship needs to be analyzed from both the customer’s and organization’s perspectives. We introduce two dimensions to define each participant’s value to the relationship. One dimension is the customer value from the customer’s perspective and the other dimension is the customer equity from the company’s perspective.

Customer value is the bundle of benefits customers perceive from a given relationship in both economic and psychological terms (Ravald and Gronroos, 1996; Gwinner *et al.*, 1998). For example, a frequent buyer program offers

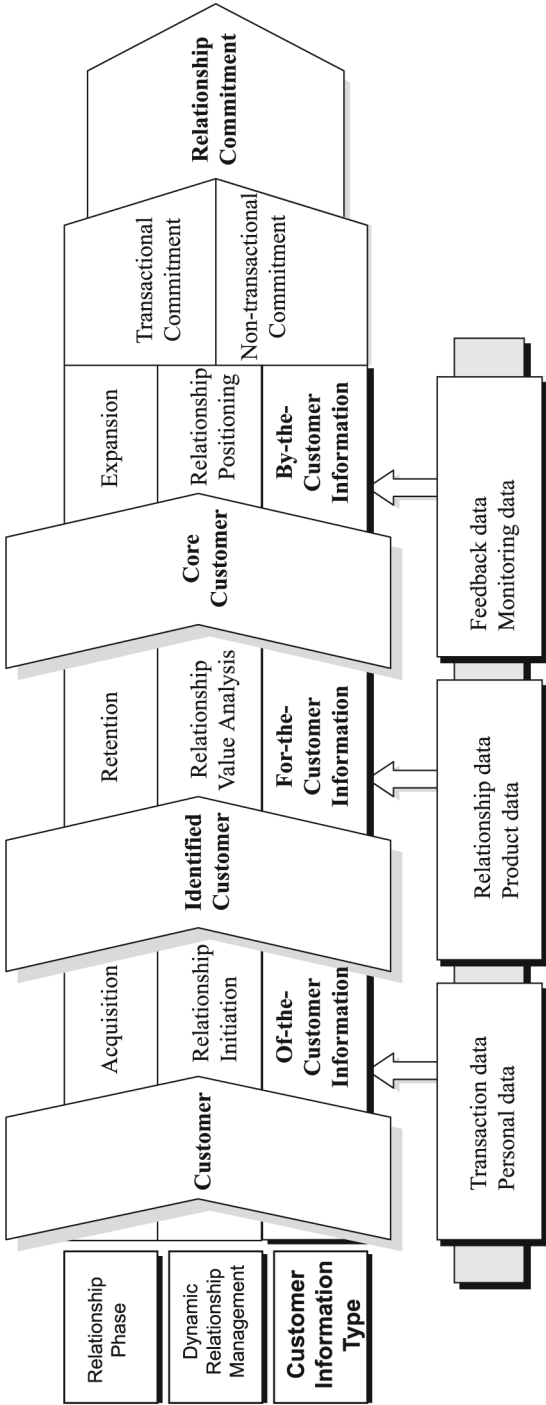


Figure 1.
A framework of dynamic
customer relationship
management

economic value such as membership discount or bonus points for repeat purchasing, while consistent communication between customers and firms creates an emotional tie between the customer and the firm by reducing perceived risks and increasing stability of the customer's future exchange with the firm. Customer equity is calculated as the sum of the customer's lifetime value (Rust *et al.*, 2000; Blattberg *et al.*, 2001). Usually, the customer's lifetime value is the profit that would have been yielded by the customer if he/she had continued purchasing for the expected number of years.

As the customer becomes more committed, he/she may create more profits by also becoming a referrer. The firms, of course, should make every effort on behalf of these customers. The equity of these customers should be much higher than others. So this type of equity – opinion leadership of core customers – is considered an important component of the customer equity.

Table I shows components and methods of evaluating the relationship values. The commitment index as part of the firm's evaluation components indicates the level of customer loyalty. It is calculated, based on factors such as relationship period, relationship intensity, and the number of referrals that a customer makes. There are many ways to increase a customer's lifetime value but they should be viewed from a long-term perspective to share the benefits with customers. To do that, companies should understand their current relationship status through analyzing mutual relationship values – customer value and customer equity. Figure 2 shows the analysis of the relationship value along the two dimensions.

The area I in Figure 2 shows that, while the company provides significant value to its customers, the equity of the customers is relatively low for the company. So, in this case, the company may be given two alternatives. First, it can expel low-equity customers by reducing the level of benefits provided. Or it can increase the customer equity by intensifying customer relationship interactions and increasing customer loyalty. The customers in area II in Figure 2 have high customer equity but their perceived customer value from the relationship is relatively low. In this case, because the current value of the relationship to the customers is low, if a competitor offers a better value proposition, they may be tempted to defect.

	Customer's view	Organization's view
Value of relationship	Customer value	Customer equity
Evaluation components	Cost of relationship (C) Extra value of relationship Economic value (EV) Socio-psychological value (SpV)	Life time value (LTV) Commitment level (CL)
Evaluation method	$(EV + SpV) - C$	$\{\sum LTV\} \times CL$

Table I.
Relationship evaluation

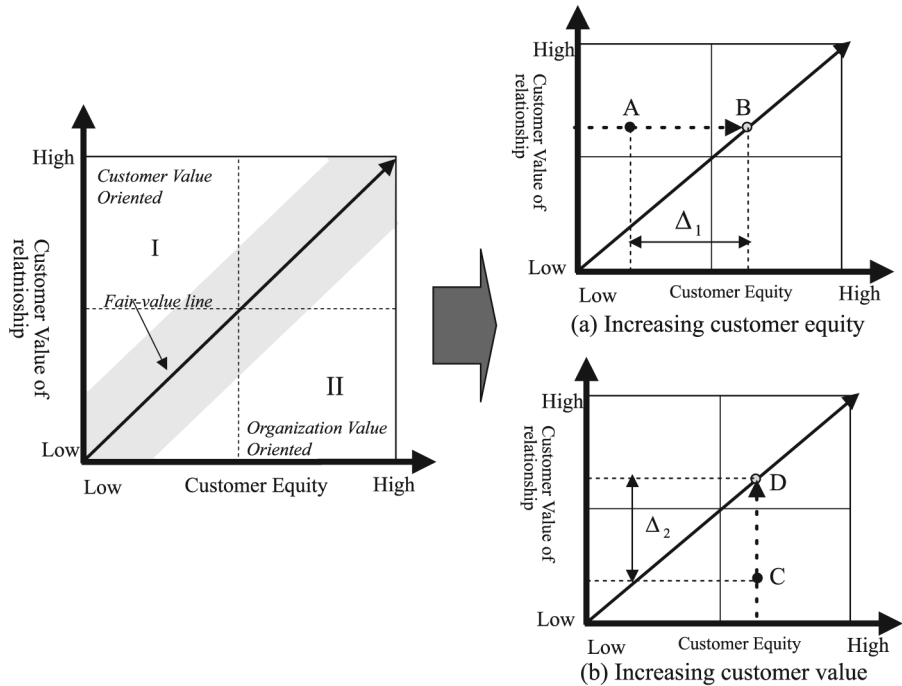


Figure 2.
Relationship value
analysis and positioning
in two dimensions

Thus when your customers (especially core customers) are located either in area I or area II of Figure 2, it suggests a state of disparity and necessitates corrective actions.

Relationship positioning

According to equity theory, when a person perceives disparity in relationship, he/she will try to change the balance in an actual or psychological manner; otherwise, he/she would attempt to end the relationship (Hatfield *et al.*, 1979). So equitable relationship building is important in the long-term relationship management. Through proper relationship positioning, it is possible to manage the customer life cycle so that firms may implement an equitable relationship strategy; by maximizing customer equity as well as enhancing the customer value.

The fair-value line is a guidance to such an equitable relationship positioning. As shown in Figure 2, if the current relationship state of an organization and its customer is located at point A, the organization does not extract fair asset value from the customer. The organization should make the relationship equitable by increasing the customer equity. According to the fair-value line, the point of equitable relationship is B, so that the effective amount that should be increased is Δ_1 . Similarly, if the point C indicates the current relationship status, the organization does not provide the customer fair

value relative to its equity, so the organization should increase its customer value by amount Δ_2 .

The shaded area on both sides of the fair value line represents the value that is mutually equitable. If the relationship is not equitable to both parties, it is difficult to maintain the relationship over the long-term. The company should periodically evaluate relationships with its customers from these perspectives, as well as from the customary cost and benefit perspective. On the fair-value line, relationship value is fair for the company and its customers but it does not guarantee that the relationship is mutually satisfactory. The ideal relationship should lead to high customer value and high customer equity compared with competitors.

Two perspectives for relationship commitment

Through proper relationship value analysis and relationship positioning, a relationship can be evolved to the level of relationship commitment that has been accepted as the ultimate level of relationship evolution (Dwyer *et al.*, 1987). The commitment concept has been identified in both cognitive and affective perspectives (Morgan and Hunt, 1994; Pritchard *et al.*, 1999). First, it is brought about by a cognitive evaluation of the value of a continued relationship with a firm. In this perspective, to induce customer commitment, firms provide customers with extra benefits and values according to their loyalty level, as judged by their repurchasing behavior. For example, to solidify their relationships with consumers, chain retailers, airlines, credit card firms have implemented frequent buyer programs that provide bonus points for large sales volumes or repeated purchasing.

We define the cognitive aspect of commitment as transactional commitment. Firms can increase their customers' transactional commitment by providing them with economic and psychological value. Economic value is added when a discount for frequent purchases or large sales volumes is offered. Psychological values can be enhanced by strengthening the convenience aspect such as offering home delivery service or individually tailored services.

The affective aspect of commitment is defined as non-transactional commitment. It grows based on trust and emotional bonding between the firm and its customers. Firms earn non-transactional customer commitment through consistent relational communication and customer participation. For example, when software products are launched, they are tested by end-users. Service providers employ customers as panels to evaluate service quality and to interview new sales persons in a recruiting process. These activities help improve organizational effectiveness and enhance customers' commitment to the firms. Today, most companies operate a Web-based homepage that makes it easy to communicate with their customers. Using this virtual space, customers make suggestions for service improvement, request after-service, or form a virtual community with common interest on the firm's products or

services. Through these activities, customers play a role as advocates or monitors for certain products and services.

In summary, for maintaining a successful long-term relationship with their customers, firms should provide differentiated relationship value and communicate continuously with their customers. Customer information lays the foundation of such sustainable relationships. With the help of information technology, organizations can manage their customer information to enhance their customers' transactional and non-transactional commitment, thereby boosting their competitiveness.

Customer information system for CRM

A customer information system (CIS) for CRM is an integrated approach for an information technology strategy to implement a relationship-oriented marketing strategy. A CIS plays a boundary-spanning role from two perspectives. First, it internally records, distributes customer information, and supports decision making for customer relationships. Customer information recorded in a database supports decision making and performance measurement by enabling the creation of customer profiles. They are used to assess customer equity so that organizations may identify their core customers (see Figure 3).

Customer equity is measured in both qualitative and quantitative ways. Quantitatively, the lifetime value of the customer can be calculated statistically by analyzing historical transaction data and predicting customers' purchasing behavior in the future. It is difficult to measure the degree of commitment by statistical techniques since the concept of commitment is not quantitative. So, it is necessary to build a rule-based model that includes objective-scale criteria on the psychological dimension of customer commitment at both the transactional and non-transactional levels. This requires data on the cognitive dimensions of customer behavior like opinion leadership, participation level, trust, etc. Since customer value of the relationship is measured from the customer's viewpoint, it is measured by analyzing the customer's feedback data.

To test whether the relationship is equitable, customer equity is compared with the customer value of the relationship relative to competitors. Normalization of each value with relative scales is required in mapping the relationship value analysis. The application program for CIS consists of rule and knowledge bases and models that support measuring the relationship value for effective customer relationship management. According to the results of the relationship value analysis, the organization can adopt a proper relationship positioning strategy aimed at creating a higher relationship value for both parties.

Second, externally, the CIS system supports the firm's interaction with its customers. Three types of customer information are organized with each other so that they are efficiently tailored to meet the customers' specific information

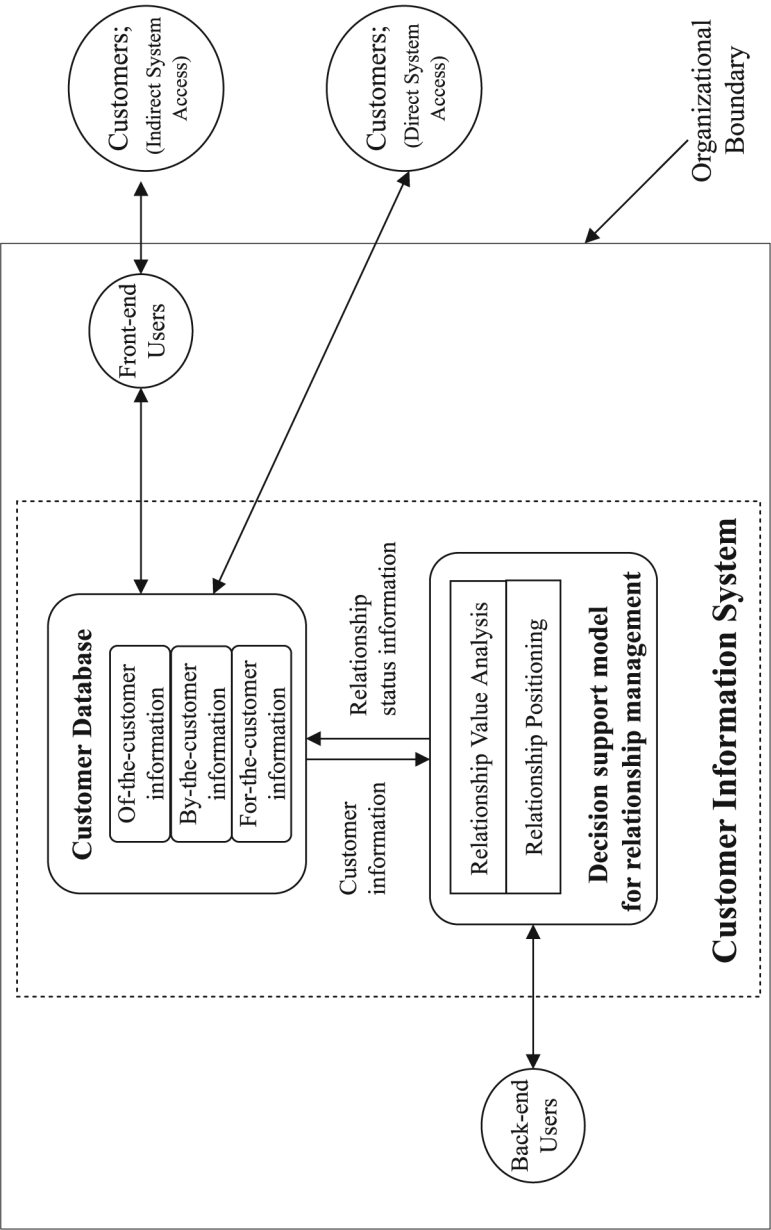


Figure 3.
Configuration of a
customer information
system for CRM

requirements. Since customer interaction may or may not be fully automated, interaction occurs in both direct and indirect system accessing. Indirect access with human interaction provides the social element of the experience so that there is an emotional element in the human dyadic. Recently, many firms give information through Web-based homepages. These homepages can provide “for-the-customer” information including company, product, and service information tailored for each customer. It also supports fully automated interactions like interactive feedback and online business services (Liu *et al.*, 1997).

The most important part of the communication strategy is to ensure that communication is sufficiently frequent. More customers are lost through a lack of follow-up, and what appears to the customer to be indifference, than through any other cause of defection. A study of competitor communications may indicate real opportunities to get closer to customers; however, since the market environment is variable, the communication frequency should be based on customer-defined expectations.

Traditional media includes both mass media (television, radio, newspaper, magazines, and direct mail) and personal communications (word-of-mouth). New forms of media encompass interactive media like videotex, CD-ROM, online services, and hypermedia computer-mediated environments. Relative to the traditional media, the new forms of media tend to afford greater consumer control and nonlinear access to more differentiated contents.

Case study: Maeil Dairy Corporation

Study approach

The central activity in doing research is to develop and test a theory. However, the framework of dynamic CRM is a new concept, and more observations are needed before a theory can be developed. Allport (1961) argues that the case study method permits the detailed observations that can reveal various nuances and subtleties of behavior that other empirical methods might miss. Similarly, Eisenhardt (1989) argues that case study research leads researchers to find new theoretical relationship and question old ones. The purpose of our case study is to explore situations and problems when the framework of dynamic CRM is applied.

In this case study, data was collected by the following steps. First, to understand the entire corporate information system environment, we referred to the enterprise information system planning reports that include overall specifications of hardware, software, human resources, and business processes, and gathered information on how the case firm started and managed their customer relationship management activities. Then we conducted several in-depth interviews with senior managers of the marketing, customer service, and information system divisions. Each interview lasted about an hour. During the first meeting, we explained our framework for CRM to each manager and

asked the interviewee to describe the related activities and programs within their division. The results of the internal customer surveys that measure customer value and perception of the products and services have been used to analyze the customer relationship management efforts from the customer's perspective (see Table II).

A framework of dynamic CRM

Case background

Maeil Dairy Corporation, a major dairy producer in Korea, was founded in 1969. Its 2001 sales was in excess of US\$500 million. While Maeil Corp. produce more than 200 products, they can be classified into two categories:

- (1) *the powdered milk product group* – which includes infant foods; and
- (2) *the liquid milk product group* – which includes milk, yogurt, juice, and mineral water.

The total sales volume of the powdered milk group represents about 30 percent of the total revenue, and the liquid milk group, 60 percent.

Maeil Corp.'s products are produced in four plants and sold through about 1,000 distributors, including agents and franchisees around the country. It has been described as a conservative and hierarchical organization, since the decision-making authority is centered at the top management.

Maeil Corp.'s customer relationship management

Maeil has managed various relationships with the consumers, including promotions and customer service. They can be classified into two types according to their product categories.

Data sources	Inputs in case study	Outputs in case study
Enterprise information system planning reports	Corporate information system architecture (H/W, S/W, human resources, business processes)	Understanding the corporate information system environment
Summary report of survey	Long-term business strategy Scope: headquarters, factory, local branch, consumers, sales persons, local agents Information system usage Product and service status	Information system supports for sales and marketing Customer perception of product and services
Interviews	Marketing manager Manager of direct mailing MIS system operators	Customer information management status CRM program and activity Customer database system configuration
Other sources	Internal reports System documents	

Table II.
Data sources, input, and output used in the case study

First, for infant food product consumers, Mothers' Membership Program has been operating since the 1970s as a sales promotion and public relations tool for consumers who have infants under the age of 12 months. This program is divided into three stages:

- (1) pregnancy;
- (2) birth to four months; and
- (3) four-12 months.

Through direct mailing (DM), the members are given educational and other promotional materials on pregnancy, birth, and infant care. Customer information is acquired from diverse channels. The major channel is the hospital where the new mother has given birth to the child. In this case, nurses in the hospital write membership initiation cards for the mothers and the DM division of the firm collects them in batches. The second channel is through women's magazines with membership application post card inserts. Hospital subscriptions account for 71.1 percent of the members, and 15.8 percent come from the magazine inserts. Maeil's customer database contains a huge amount of "of-the-customer" information – 460,000 records out of the 600,000 women who deliver babies in Korea each year, but it is mostly profile type data. Since there is no customer transaction data recorded in the database, Maeil Corp. cannot identify their real customers, even with the 460,000 pieces of personal data. The data is not available to the market research department. Recently, Maeil Corp. attempted to sell infant food to members at discount prices through direct mailing. But the response rate was only 0.96 percent, which is low relative to the usual return rate of direct marketing in Korea (about 1.5 percent).

For-the-customer information, like product and service information, membership information, and other knowledge about infant care, is periodically provided through mail or sales promotions. But customized information for specific requests from individual customers is not provided. Although four or five infant-care consultants are working at the local branches, individual customer consultation information is not stored in the organizational memory.

Although Maeil Corp. has data on 76 percent of the total infant food consumers, it is not made available for the market research and decision-making. Because of the customer privacy concern, the system and data are restricted to DM operators only. Since there is no behavioral and psychographical data, it is impossible to apply the information to market research and decision-making at this point.

In addition, data accuracy is a basic problem in data collection and manipulation. Most of the membership data is acquired in the hospital. However, membership subscription is not initiated by the mothers but by a nurse. Consequently, mothers often do not know of their membership to Maeil, and do not care about the accuracy of the data. This is the reason why

the DM response rate was low relative to the average response rate of the local direct marketing.

Except for infant food products, Maeil Corp. does not manage their end consumers directly. Access to customer information is blocked by the distributors such as franchisees and sales agents across the country. No information is collected about the distributors' customers like retailers, end consumers, and convenience stores. For instance, since the information on home-delivery consumers is kept within the delivery person's memory, when he or she retires, the information disappears.

Approximately, 25 percent of the total consumers prefer home-delivery of milk; the others purchase it at the retailers. Data from an internal survey indicates that the reason for home-delivery service is convenience (55.7 percent) and freshness of the milk (37.3 percent). A total of 89 percent of customers are willing to continue to have the service delivery. Recently, however, the number of home-delivery consumers has decreased, as many of them opt to purchase milk at the retail stores. The reason for this change is that consumers are putting more value on price than on convenience. Consequently, the transaction points of consumers in the distribution channel are moving from distributors to small or large-scale retailers.

Relationship value analysis and positioning

We can analyze the relationship value of the two different customer relationships only qualitatively since Maeil Corp. lacks transactional data about customers and there are no formal channels for monitoring their customer value (see Table III).

In the Mothers' Membership Program, customers perceive a low economic value and high psychological value. They strongly want to learn how to care for themselves during their pregnancy and about infant care. Through

Relationship	Value	Mothers' membership	Home-delivery consumers
Customer value	Economic value	Low economic value; partially, direct sales with discounted price	No economic value
	Psychological value	Valuable knowledge about pregnancy and infant care	Convenience in purchasing Freshness of products
Customer equity	Customer LTV	Low life time value Short relationship periods	High lifetime value Long relationship periods
	Degree of commitment	Low commitment Non-transactional commitment in some members who monitor products and services	High degree of commitment Transactional commitment

Table III.
Qualitative evaluation
of customer relationship

campaigns with expert lecturers in related areas and direct mailing of informational materials, Maeil satisfies the customers' needs and demands. Some of them communicate with the central research institute individually. They monitor products and services through personal communication channels, so they are committed non-transactionally by participating in various firm-sponsored activities. However, since there is no data about individual customer behavior, Maeil Corp. cannot manage its customer relationships systematically; it is not easy to estimate the relationship value, identify core customers, or, by relationship positioning strategy, create more core customers from standard customers.

The customer equity in Maeil Corp. Mothers' Membership is low because the relationship time is short – less than 12 months. However, home-delivery consumers have long-term relationships relative to non-home-delivery consumers and most of them are willing to retain this relationship in the future. They are highly committed transactionally to Maeil Corp.

While the members of Maeil Corp. Mothers' Membership give high value to this relationship, there is no difference between members and non-members in customer equity since the degree of transactional commitment is low and the relationship period is short. Home-delivery consumers show high equity since they are committed transactionally and for a long time. However, there is no economic benefit and they perceive no values other than convenience and freshness, so they are beginning to purchase through other distribution channels like supermarkets and convenience stores. Eventually, the number of home-delivery consumers will decrease and their equity will be reduced.

From the relationship positioning perspective, Maeil Corp. needs to raise the equity level of its Mothers' Membership Program by tracking the members' sales transaction data. On the other hand, it should provide higher value to its home-delivery milk customers by tracking their personal information.

Customer information system configuration

Maeil Corp. is in the process of building a customer information system that provides the three types of customer information. Until now, their marketing strategy has been product-oriented, not customer-oriented. There are product managers but no customer managers. Currently, the marketing function is divided into two categories – liquid milk products and powdered milk products – so the company cannot adequately manage cross-selling among its products because of the lack of information sharing.

Maeil Corp. should build a long-term customer relationship strategy that makes the product-dependent customers into life-time customers. The two customer relationships should be integrated into one and managed by customer managers, not by product managers. It is possible to keep customers “from the cradle to the grave” through careful and integrated customer life cycle management. Since Maeil Corp.'s products are purchased for the purpose of

health-care and end consumers are diverse in age, it would be feasible to develop product packages with tailored information and health-care instructions that are provided to individual customers or families according to their age.

For relationship initiation through mutual information sharing, various contact points with customers should be established. The personal and transaction data should be collected through direct and indirect contact points. As direct contact points, available contact points are PC-based VAN services, Internet homepages, and direct mailings associated with customer databases. It is also possible to use the POS data from its affiliated convenience store chain (LOVE M) and agents. Maeil Corp. needs to design a strategic rewards system that stimulates customers to initiate the relationship by giving their private information voluntarily. Coupon redemption is a typical tool for acquiring consumer information. Consumers who want to initiate a relationship with Maeil Corp. give their information with collected coupons that show their transaction history.

The relationship value analysis needs to be conducted periodically by measuring customer value and equity. A relationship positioning strategy and follow-up procedures also need to be established. For the internal use of customer information, the Central Research Lab and the Marketing Division can share customer information for their own purposes and still protect the privacy of customers.

Through the integrated CIS as shown in Figure 4, Maeil Corp. can plan to give life-time health care and dairy food services to their customers. Packaged food products, for health care information and value-added services can be provided for individual customers at each customer life cycle stage.

Conclusion and limitation

We have defined the concept of “identified” and “core” customers from the information processing perspective. In our definition, it is impossible to manage relationships with unidentified customers even though they are committed to your organization. Based on these definitions, we proposed a framework of dynamic customer relationship management, consisting of three stages: first, a customer acquisition stage, where the organization recognizes unidentified consumers as customers who are identifiable by collecting the information about them through diverse communication channels; second, a customer retention stage, where the organization builds equitable relationships through relationship value analysis so that ordinary customers turn into core customers who are committed to the relationship with high equity; finally, a customer expansion stage, in that core customers play the role of expanding the company’s customer base through word of mouth marketing and active interactions with the company. Throughout the entire customer lifecycle, an

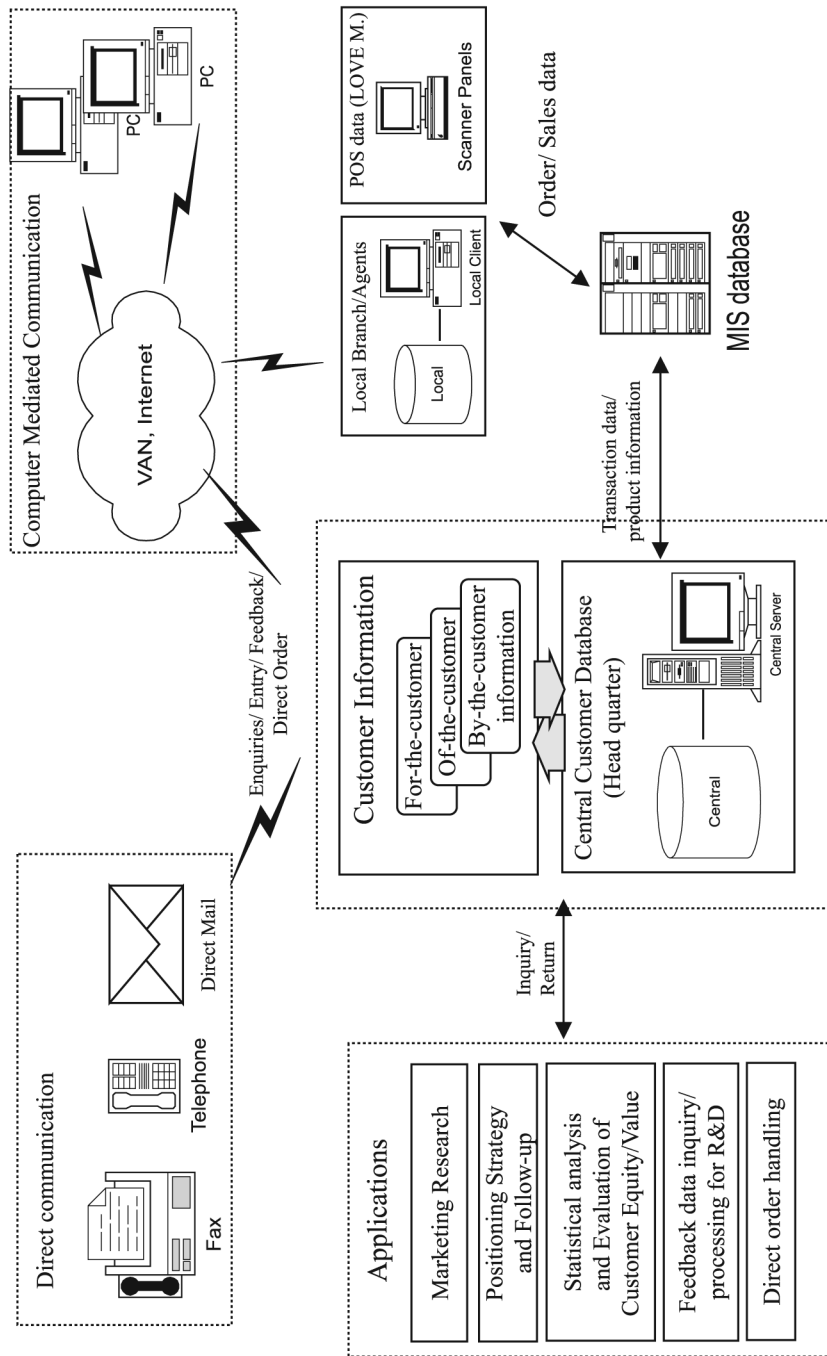


Figure 4.
Configuration of
customer information
system for Mael Corp.

organization is expected to measure relationship value from both perspectives and build a proper relationship positioning strategy.

Once a proper strategy is chosen, in order to collect and analyze the required customer information, it is necessary to build a CIS customer for relationship management. A CIS for relationship management consists of a customer database, a communication channel, and an application for relationship management. Such CIS should support all three types of customer information: of-the-customer, for-the-customer, and by-the-customer information. Through the framework of dynamic customer relationship management and CIS, organizations should be able to manage the entire customer life cycle and make one-time buyers into lifetime core customers.

However, we have studied the customer relationship management in a qualitative manner so the theoretical framework proposed may need further validation quantitatively based on real world CRM implementation data. The concept of customer value and equity as well as CIS architecture should be refined with detailed criteria to be applicable to other industries. In the relationship value analysis, measuring customer equity is not easy. In practice, the customer equity has been calculated by subtracting the customer management costs from the customer lifetime value, which is the present value of the future potential profitability. Using statistical techniques and prediction models, it can be estimated from the firm's transaction database but, with this method, only the behavioral aspect is considered. Socio-psychological factors like degree of commitment, as we indicated in this paper, should also be included in estimating more relevant customer equity and developing effective CRM strategies.

References

- Allport, G.W. (1961), *Patterns in Growth and Personality*, Holt, Rinehart & Winston, New York, NY.
- Barnes, J.G. (1994), "Close to the customer: but is it really a relationship?", *Journal of Marketing Management*, Vol. 10, pp. 561-70.
- Barnes, J.G. (1995), "Establishing relationships – 0 getting closer to the customer may be more difficult than you think", *Irish Marketing Review*, Vol. 8, pp. 107-16.
- Berry, L.L. (1983), "Relationship marketing", in Berry, L.L., Shostack, G.L. and Upah, G.D. (Eds), *Perspectives on Services Marketing*, American Marketing Association, Chicago, IL, pp. 25-8.
- Blattberg, R.C. and Deighton, J. (1996), "Manage marketing by the customer equity test", *Harvard Business Review*, July/August, pp. 136-44.
- Blattberg, R.C., Getz, G. and Thomas, J.S. (2001), *Customer Equity: Building and Managing Relationships as Valuable Assets*, Harvard Business School Press, Boston, MA.
- Copulsky, J.R. and Wolf, M.J. (1990), "Relationship marketing: positioning for the future", *Journal of Business Strategy*, July/August, pp. 16-20.
- Duncan, T. and Moriarty, S.E. (1998), "A communication-based marketing model for managing relationships", *Journal of Marketing*, Vol. 62, pp. 1-13.

- Dwyer, F.R., Schurr, P.H. and Oh, S. (1987), "Developing buyer-seller relationships", *Journal of Marketing*, Vol. 51 April, pp. 11-27.
- Eisenhardt, K.M. (1989), "Building theories from case study research", *Academy of Management Review*, Vol. 14, pp. 532-50.
- El Sawy, O.A. and Bowles, G. (1997), "Redesigning the customer support process for the electronic economy: insights from storage dimensions", *MIS Quarterly*, Vol. 21 No. 4, pp. 457-83.
- Goodhue, D.L., Wybo, M.D. and Kirsch, L.J. (1992), "The impact of data integration on the costs and benefits of information systems", *MIS Quarterly*, Vol. 16 No. 3, pp. 11-34.
- Gwinner, K.P., Gremmler, D.D. and Bitner, M.J. (1998), "Relational benefits in services industries: the customer's perspective", *Journal of the Academy of Marketing Science*, Vol. 26 No. 2, pp. 101-14.
- Hatfield, E., Utne, M.K. and Trarupmann, J. (1979), "Equity theory and intimate relationships", in Burges, R.L. and Huston, T.L. (Eds), *Social Exchange in Developing Relationships*, Academic Press, New York, NY.
- Jutla, D., Craig, J. and Bodorik, P. (2001), "Enabling and measuring electronic customer relationship management readiness", in *Proceedings of the 34th Hawaii International Conference on System Science*.
- Kohli, R., Piontek, F., Ellington, T., VanOsdol, T., Shepard, M. and Brazel, G. (2001), "Managing customer relationship through e-business decision support applications: a case of hospital-physician collaboration", *Decision Support Systems*, Vol. 32, pp. 171-87.
- Liu, C., Arnett, K.P., Capella, L.M. and Beatty, R.C. (1997), "Web sites of the *Fortune* 500 companies: facing customers through home pages", *Information and Management*, Vol. 31 No. 6, pp. 335-45.
- Mentzas, G.N. (1997), "Re-engineering banking with object-oriented models: towards customer information systems", *International Journal of Information Management*, Vol. 17 No. 3, pp. 179-97.
- Morgan, R.M. and Hunt, S.D. (1994), "The commitment-trust theory of relationship marketing", *Journal of Marketing*, Vol. 58 July, pp. 20-38.
- O'Malley, L. and Tynan, C. (2000), "Relationship marketing in consumer markets: rhetoric or reality?", *European Journal of Marketing*, Vol. 34 No. 1, pp. 797-815.
- Palmer, A.J. (1995), "Relationship marketing: local implementation of a universal concept", *International Business Review*, Vol. 4 No. 4, pp. 471-81.
- Peppers, D. and Rogers, M. (1995), "A new marketing paradigm: share of customer, not market share", *Planning Review*, March/April, pp. 14-18.
- Pritchard, M.P., Havitz, M.E. and Howard, D.R. (1999), "Analyzing the commitment-loyalty link in service contexts", *Journal of the Academy of Marketing Science*, Vol. 27 No. 3, pp. 333-48.
- Rangaswamy, A. (1993), "Marketing decision models: from linear programs to knowledge-based systems", *Hand Books in OR & MS*, Vol. 5, pp. 733-71.
- Ravald, A. and Gronroos, C. (1996), "The value concept and relationship marketing", *European Journal of Marketing*, Vol. 30 No. 2, pp. 19-30.
- Reichheld, F.F. and Sasser, W.E. Jr (1990), "Zero defections: quality comes to services", *Harvard Business Review*, September/October, pp. 105-11.
- Rigby, D.K., Reichheld, F.F. and Scheffer, P. (2002), "Avoid the four perils of CRM", *Harvard Business Review*, February, pp. 101-9.

-
- Rowe, G.W. and Barnes, J.G. (1998), "Relationship marketing and sustained competitive advantage", *Journal of Market Focused Management*, Vol. 2, pp. 281-97.
- Rust, R., Zeithaml, V.A. and Lemon, K.N. (2000), *Driving Customer Equity*, The Free Press, New York, NY.
- Sharma, N. and Patterson, P.G. (1999), "The impact of communication effectiveness and service quality on relationship commitment in consumer, professional services", *The Journal of Service Marketing*, Vol. 13 No. 2, pp. 151-70.
- Treacy, M. and Wiersema, F. (1993), "Customer intimacy and other value disciplines", *Harvard Business Review*, Vol. 71 No. 1, pp. 84-93.
- Turnbull, P.W. and Wilson, D.T. (1989), "Developing and protecting profitable customer relationships", *Industrial Marketing Management*, Vol. 18, pp. 233-8.
- Wells, J.D., Fuerst, W.L. and Choobineh, J. (1999), "Managing information technology (IT) for one-to-one customer interaction", *Information and Management*, Vol. 35, pp. 53-62.
- Winer, R.S. (2001), "A framework for customer relationship management", *California Management Review*, Vol. 43 No. 4, pp. 89-105.
- Woodcock, N. and Starkey, M. (2001), "'I wouldn't start from here': finding a way in CRM projects", *Journal of Database Marketing*, Vol. 18 No. 1, pp. 61-74.